

FTK Imager and Autopsy



Scan me

Disk Imaging and Artifact Recovery in Autopsy

Terminology:

Image : copying of a flash drive that has information

Acquisition: The viewing of the image from the flash in order to gather information

Data compression: when the information from the flash drive is compressed together as a file to save up space

Verification: The information on the flash drive which was imaged maintains its integrity

Chain of custody: is the record of who handled the evidence when they handled it, and what was done with it.

Description:

FTK Imager is a specialized digital forensics utility designed to capture exact, bit-level copies of storage media while maintaining the original data's integrity. It enables investigators to preview files, generate verification hashes, and create forensic images in formats such as RAW, E01, and AD1, making it a core tool for reliable evidence acquisition in forensic investigations.

Process:

1. Identification
2. Preservation
3. Acquisition
4. Examination
5. Analysis
6. Documentation
7. Reporting
8. Presentation.

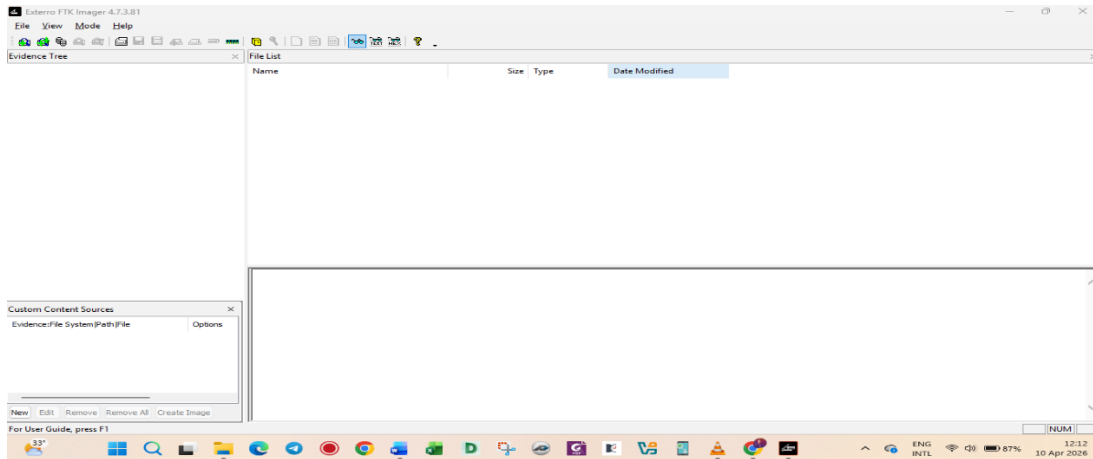
Tools used: FTK Imager and Autopsy

Implementation:

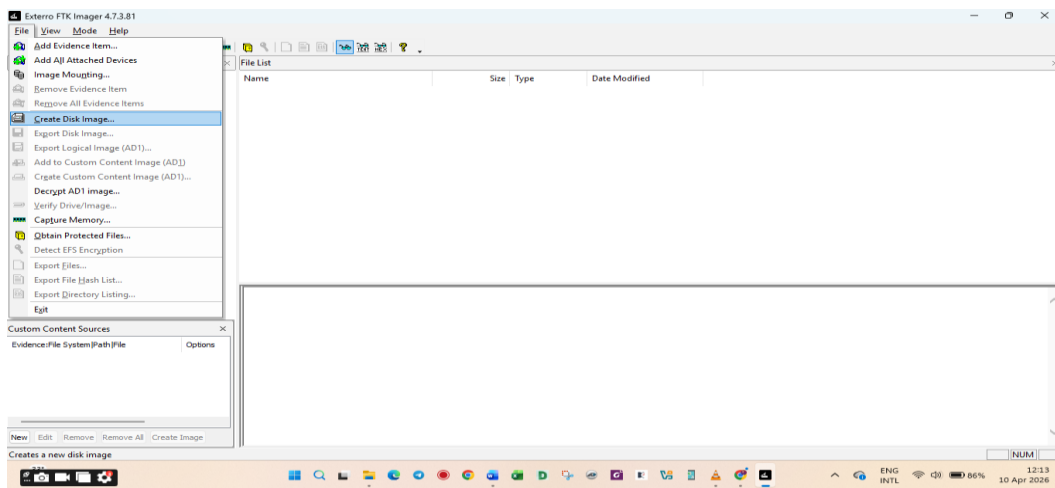
1. Acquire volume image (by creating a disk image of the device)
2. Read volume image (using the Add evidence)
3. Mounting image mode

Purpose of this project: To recover deleted files from a flash drive

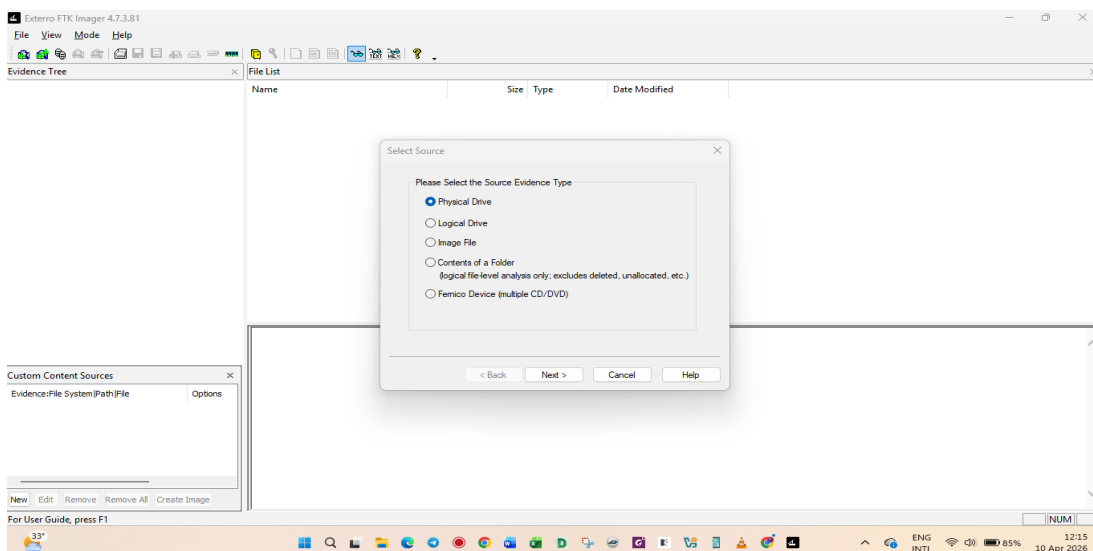
Acquire the volume image (creating a disk image of the evidence)



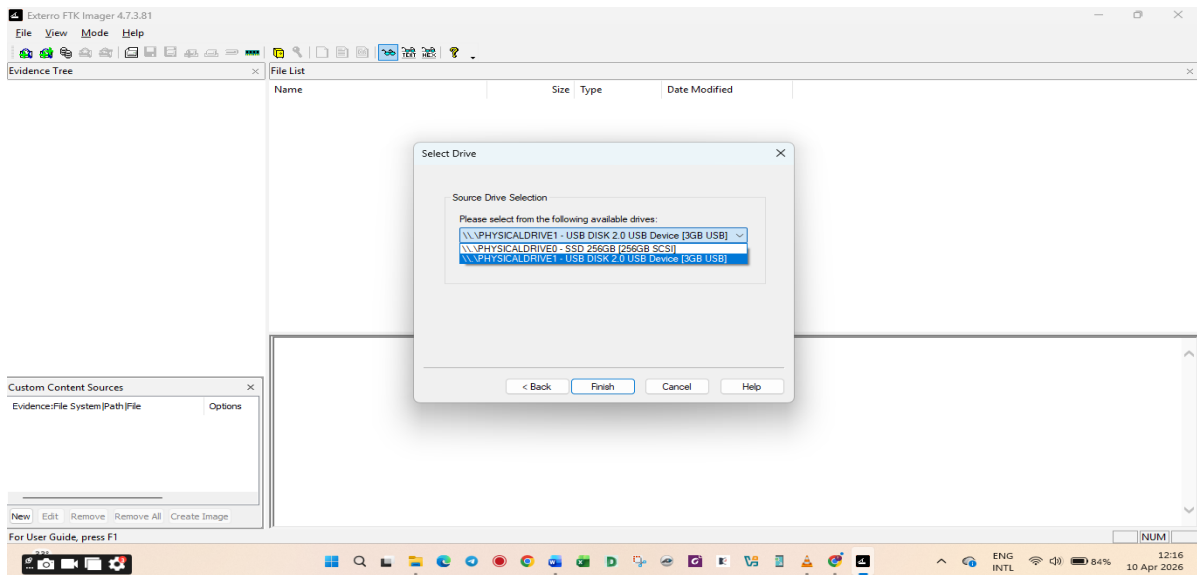
Launching of the FTK Imager



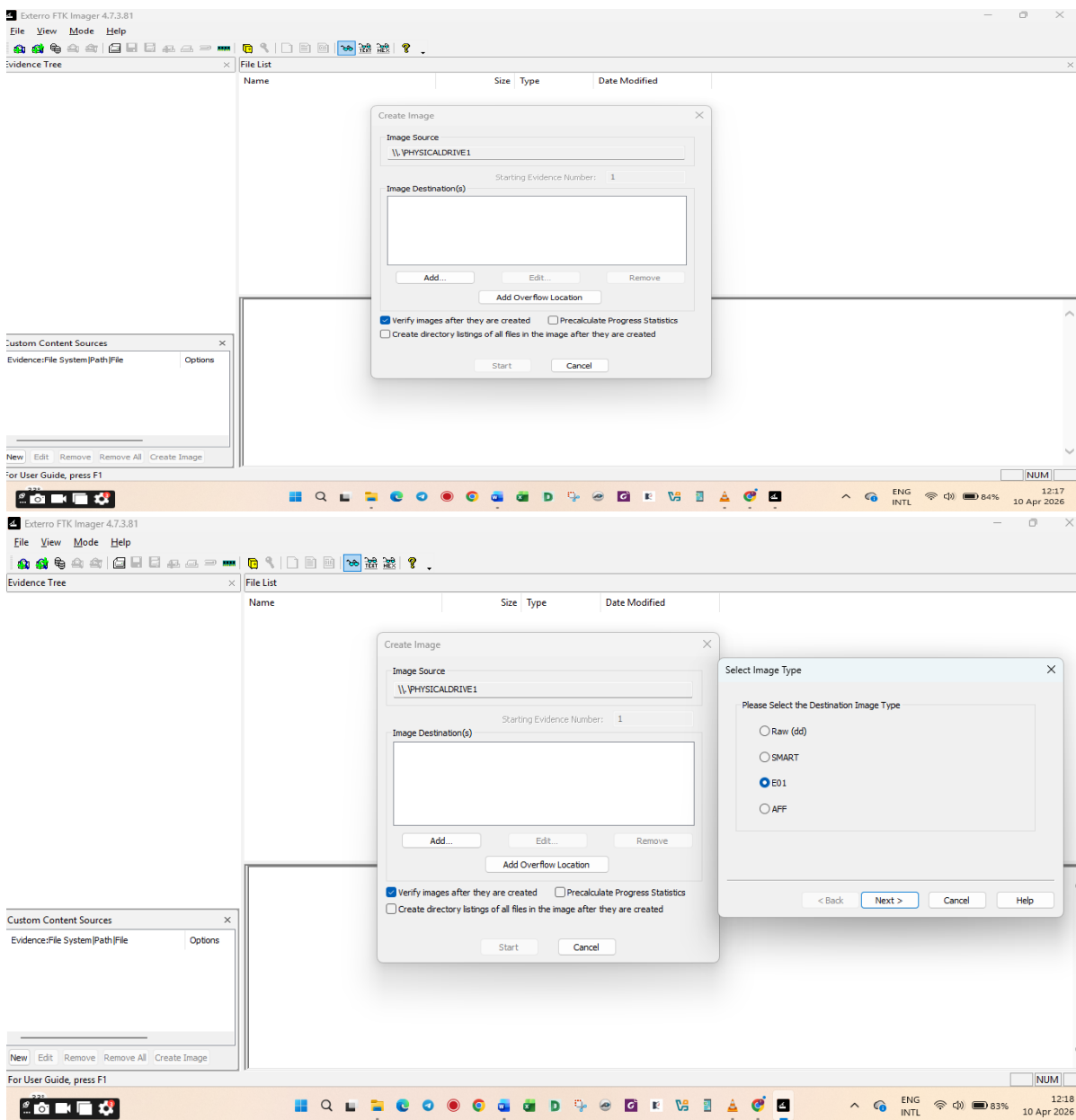
creation of disk image



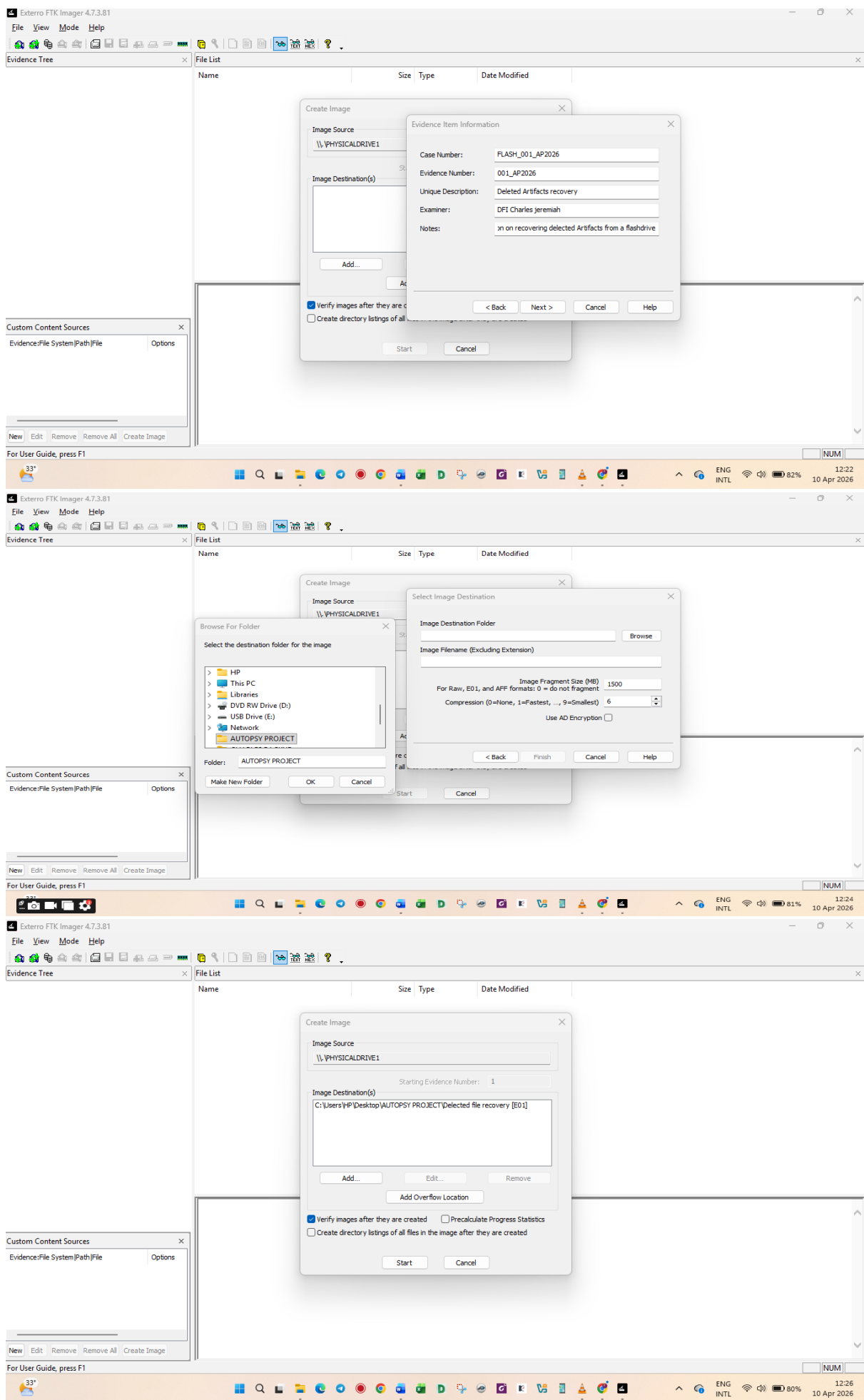
Selecting the device plugged in which is a physical device (Flash drive)



Selecting the name of the physical device plugged in



selecting the image type as E01




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Created By Exterro® FTK® Imager 4.7.3.81

Case Information:
Acquired using: ADI4.7.3.81
Case Number: FLASH_001_AP2026
Evidence Number: 001_AP2026
Unique description: Deleted Artifacts recovery
Examiner: DFI Charles Jeremiah
Notes: Simulation on recovering deleted Artifacts from a flashdrive

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Information for C:\Users\HP\Desktop\AUTOPSY PROJECT\Deleted file recovery:

Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 471
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 7,581,696
[Physical Drive Information]
Drive Model: USB DISK 2.0 USB Device
Drive Serial Number: 0C7C17C53090
Drive Interface Type: USB
Removable drive: True
Source data size: 3702 MB
Sector count: 7581696
[Computed Hashes]
MD5 checksum: fa8fa7fcf09e6c6a61ad6bbc24c3a7e2
SHA1 checksum: 02f215a0a801b01ccabfe7bfd0164f200ff88497

Image Information:
Acquisition started: Fri Apr 10 12:27:53 2026
Acquisition finished: Fri Apr 10 12:29:48 2026
Segment list:
C:\Users\HP\Desktop\AUTOPSY PROJECT\Deleted file recovery.E01
C:\Users\HP\Desktop\AUTOPSY PROJECT\Deleted file recovery.E02
COMPUTED HASH : fa8fa7fcf09e6c6a61ad6bbc24c3a7e2
COMPUTED HASH : 02f215a0a801b01ccabfe7bfd0164f200ff88497

Image Verification Results:
Verification started: Fri Apr 10 12:29:48 2026
Verification finished: Fri Apr 10 12:30:40 2026
MD5 checksum: fa8fa7fcf09e6c6a61ad6bbc24c3a7e2 : verified
SHA1 checksum: 02f215a0a801b01ccabfe7bfd0164f200ff88497 : verified
|

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General report generated after creating a copy of the evidence file(Flash drive)

A report was generated from FTK Imager following the creation of a forensic image of the flash-drive evidence. The process included successful hash verification to confirm that the acquired image remained unaltered. At this stage of the investigation, the first four phases Identification, Preservation, Acquisition, and Examination have been fully completed.

Autopsy

Purpose: Analysis

Description :

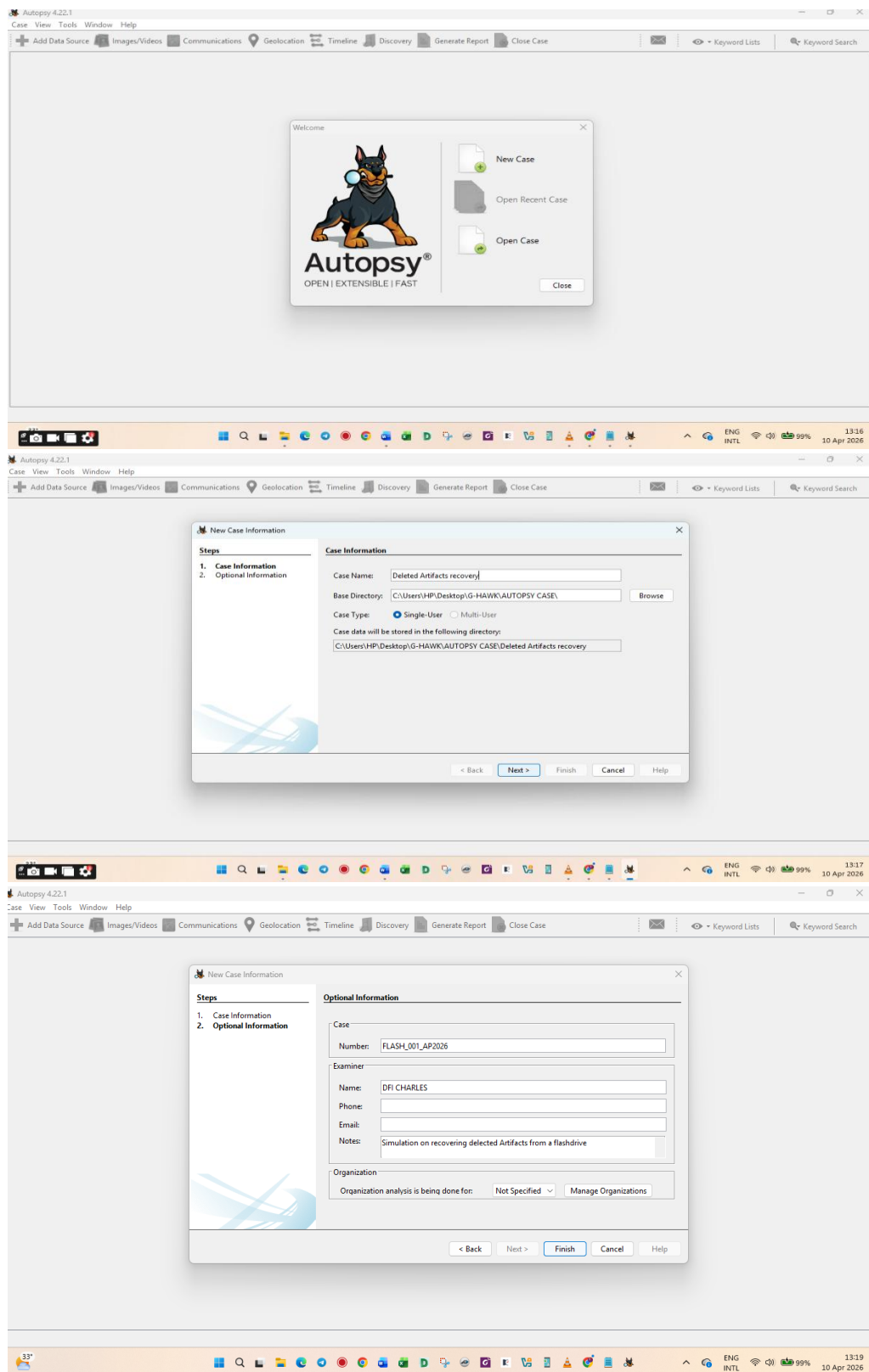
Autopsy is a digital forensics analysis tool that helps investigators examine forensic images and uncover evidence such as files, metadata, deleted items, and user activity.

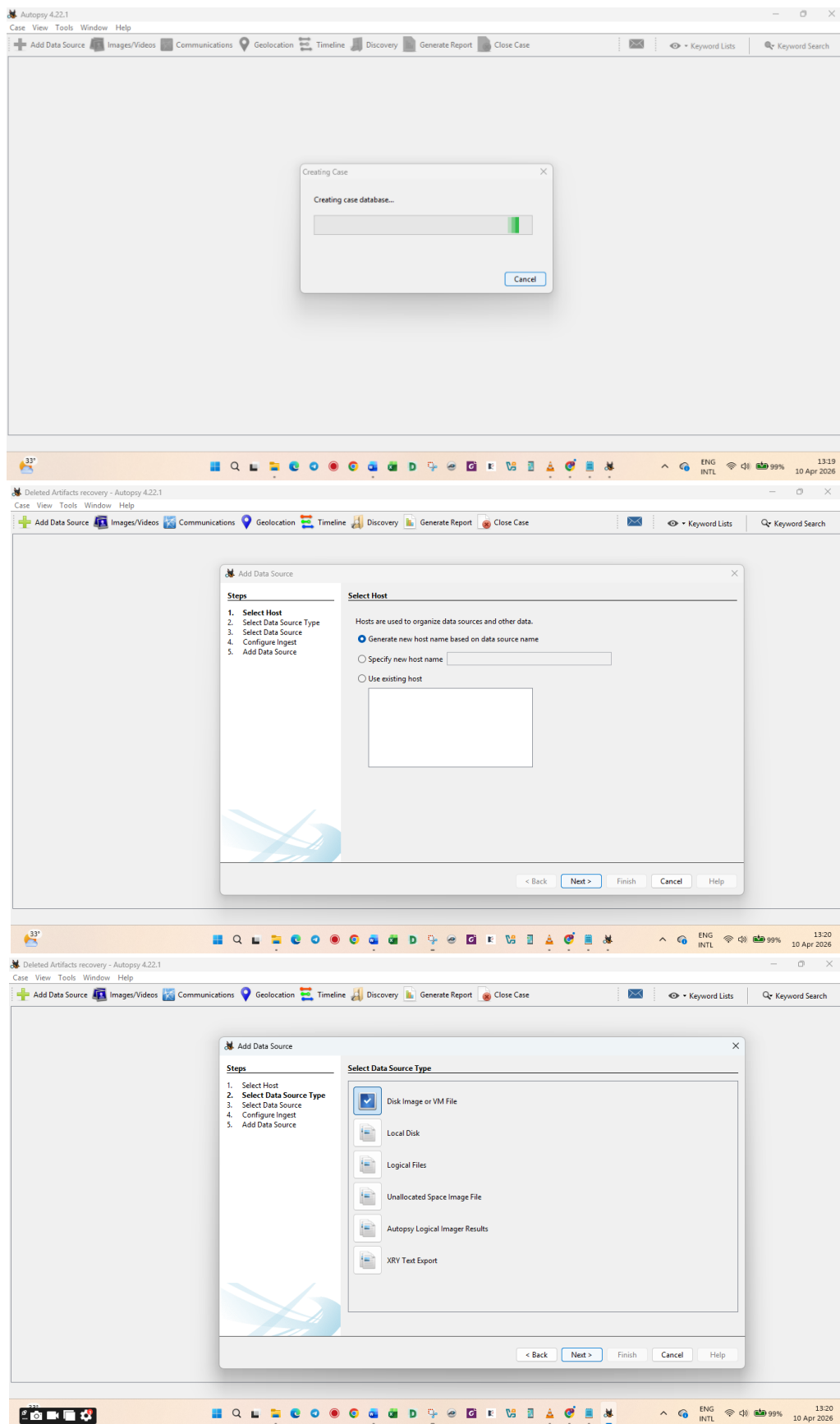
Process:

1. Case Creation
2. Adding the Data Source
3. Running Ingest Modules
4. Examining Extracted Artifacts
5. Generating the Forensic Report

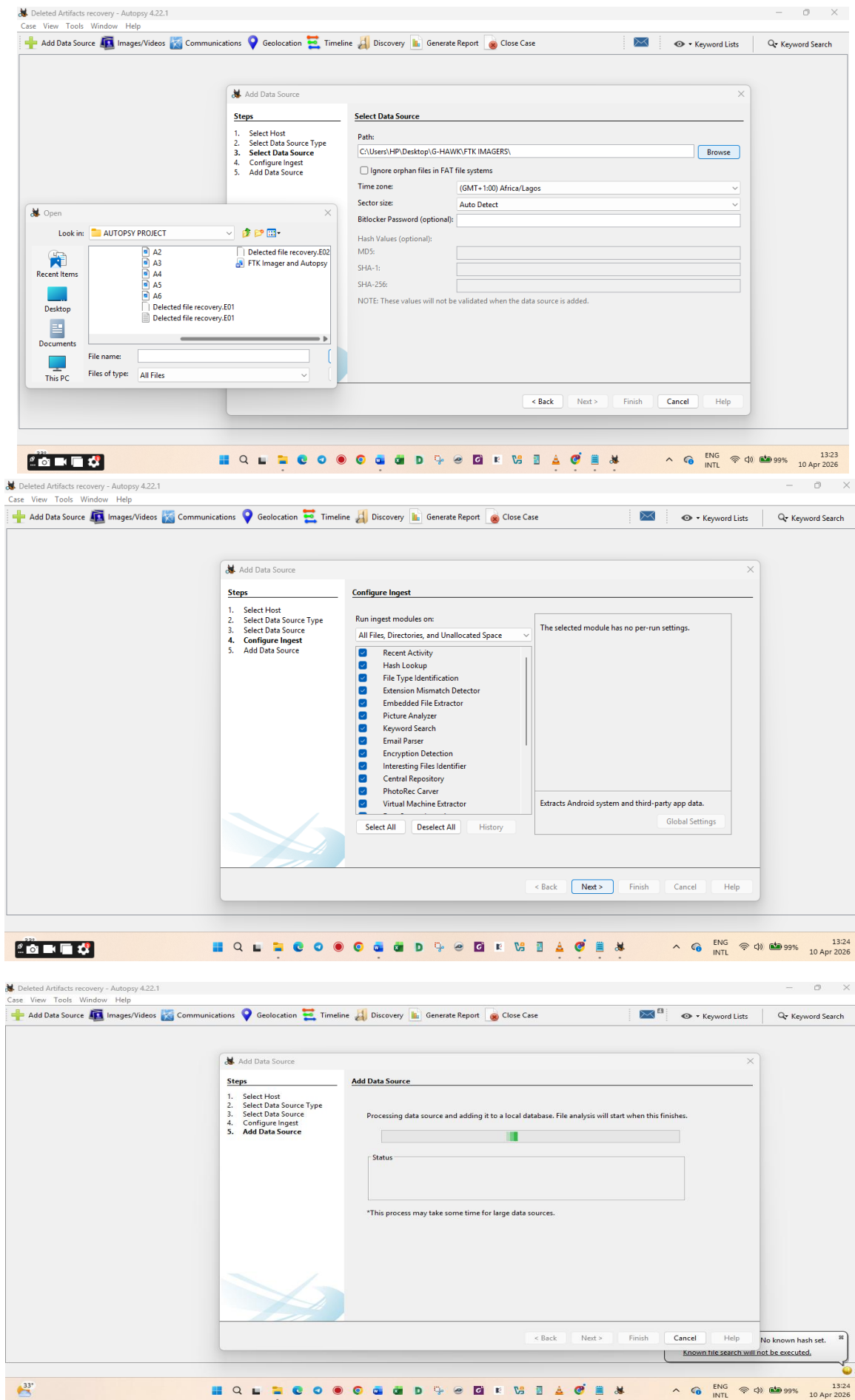
Implementation:

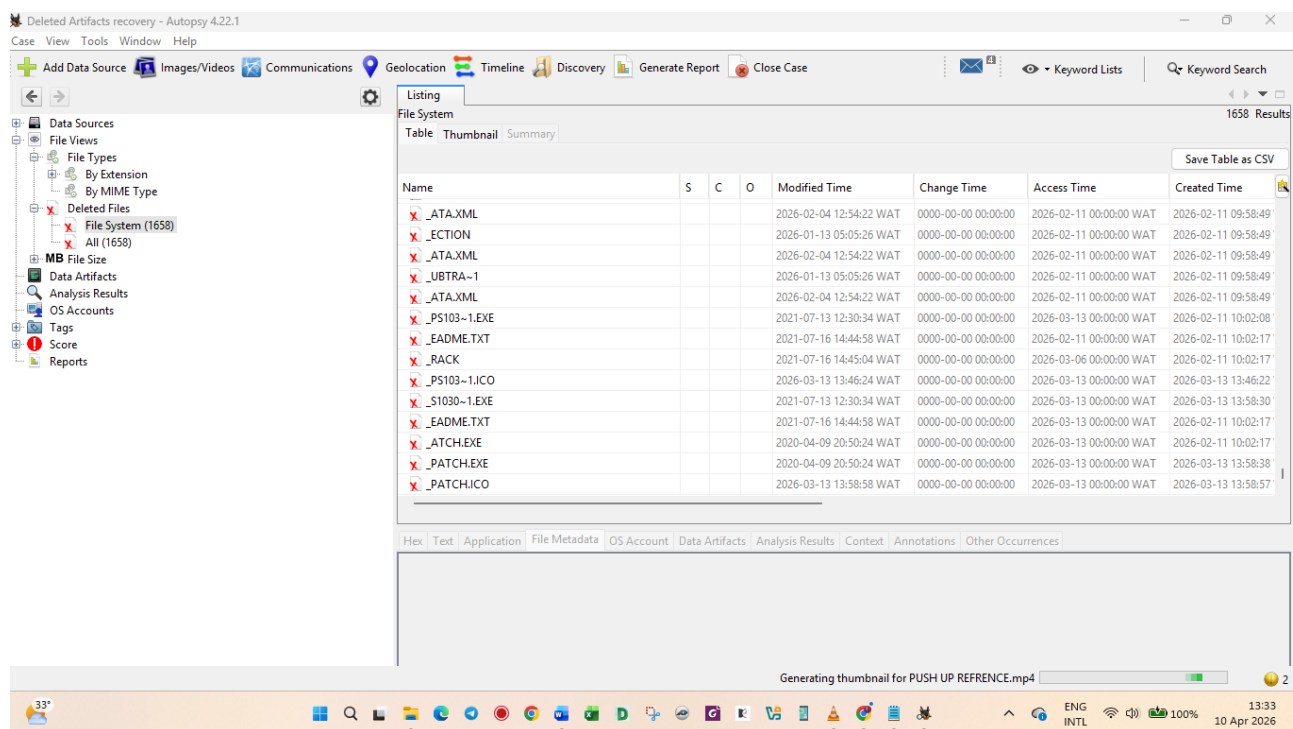
1. Create a new case and enter all required case details
2. Add the forensic image or device as a data source
3. Select and configure ingest modules needed for the investigation
4. Process and analyze extracted artifacts using Autopsy's interface
5. Generate and export a forensic report summarizing findings





Ingestion of E01





After completing the core Autopsy processes Case Creation, Adding the Data Source, Running Ingest Modules, and Examining Extracted Artifacts the E01 image generated with FTK Imager was successfully analyzed. The examination revealed documents dating from 2020 to 2026, indicating the period during which the flash drive had been in use. Autopsy also identified a significant number of deleted files on the device. This confirms that the remaining investigative phases Analysis, Documentation, Reporting, and Presentation were effectively carried out.

The investigation was successful, with the analysis showing that 1,658 files had been deleted and were recoverable by extracting the required items to a secure workstation. Overall, the process demonstrated a clear and well-maintained chain of custody throughout the hands-on exercise.